PSEUDO-COHERENT OBJECTS OF A DERIVED COMPLETE RING ARE DERIVED COMPLETE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Sub-quotients of derived complete modules are derived complete, so the cohomology groups of a complex of finite free modules over a derived complete ring are derived complete. invoked at proof lines 26137-26139. Makes each H^n(K) derived complete.
- A complex whose cohomology groups are all derived complete is derived complete. invoked at proof lines 26139-26140. Lifts the cohomological completeness to K itself.

Source reference. Original lemma 091R: sources/stacks-more-algebra/pinned-original-source.tex, source lines 25963–25963. Cited at proof lines 26132-26133 to explain the parenthetical hypothesis; the conclusion is used inside the counted span.

Source reference. Original lemma 091U: sources/stacks-more-algebra/pinned-original-source.tex, source lines 26068–26068. Cited at proof lines 26138-26140 in parts (1) and (2) to conclude each cohomology group and then K itself is derived complete; the conclusion is used inside the counted span.

Lemma 0.2. *Let A be a ring and $I \subset A$ an ideal. If A is derived complete (eg. I-adically complete) then any pseudo-coherent object of $D(A)$ is derived complete.*

Proof. (Lemma 091R explains the parenthetical statement of the lemma.) Let K be a pseudo-coherent object of $D(A)$. By definition this means K is represented by a bounded above complex $K^\bullet$ of finite free A-modules. Since A is derived complete it follows that $H^n(K)$ is derived complete for all n, by part (1) of Lemma 091U. This in turn implies that K is derived complete by part (2) of the same lemma. □

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